

Part 5: Addressing Modes

1. **What is immediate addressing? Give an example.** The data (operand) is part of the instruction itself. Example: `MOV AX, 5` (5 is the immediate value).
2. **What is register addressing? Give an example.** The data is stored in a register. Example: `MOV AX, BX` (moves data from BX to AX).
3. **What is direct addressing?** You provide the exact memory address (or variable name) where the data is stored. Example: `MOV AX, [1234h]`.
4. **Explain indirect addressing mode.** The memory address is not given directly but is stored inside a register (like BX, SI, or DI).
5. **What is register indirect addressing?** You access memory using an address held in a base or index register. Example: `MOV AX, [BX]`.
6. **What is based addressing mode?** Uses a base register (BX or BP) plus a displacement (offset) to calculate the address. Useful for accessing data structures.
7. **What is indexed addressing mode?** Uses an index register (SI or DI) plus a displacement. Useful for accessing array elements.
8. **What is based plus index addressing mode?** Combines a base register (BX/BP) and an index register (SI/DI) to calculate the address. Example: `MOV AX, [BX+SI]`.
9. **How do you access memory using SI and DI?** Put the address in the register and use brackets. Example: `MOV AL, [SI]` moves the byte at the address in SI to AL.
10. **Which addressing mode is fastest? Register addressing** is fastest because the data is already inside the CPU and no memory access is required.
11. **How is segment:offset used in addressing?** The physical address is calculated as `Segment * 10h + Offset`. The segment register (like DS) provides the base, and the instruction provides the offset.
12. **Explain relative addressing.** Used in jump instructions (like `JMP`). The operand is an offset relative to the current Instruction Pointer (IP) location.
13. **How do you calculate effective address in 8086?** Effective Address (EA) = Base + Index + Displacement. The CPU calculates this automatically based on the addressing mode used.
14. **Which addressing mode is used for arrays? Indexed addressing** (using SI or DI) is typically used to iterate through arrays.
15. **Can immediate data be used in memory operations?** Yes, but you must specify the size. Example: `MOV [BX], 5` is invalid, but `MOV BYTE PTR [BX], 5` is valid.